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SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME:

Cloud Computing and
Big Data Analytics

COURSE CODE:

CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I YOGA SHREE G (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. ygg

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Cloud Computing and Big Data Analytics.

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Reg No: 192573022.

AT1 - Short Answer.

- 1) Describe the evolution from standalone computing to cloud based systems. Summarize key technological transitions that enabled cloud computing.

Standalone computers worked independently with local storage and software.

Client-server computing enabled resource sharing over networks. The Internet connected systems globally.

Virtualization allowed multiple virtual machines on one server. Cloud computing provides on demand computing resources over the Internet.

- 2) Hypervisors in Virtualization.

A hypervisor creates and manages virtual machines.

Type 1 Hypervisor: Runs directly on hardware; faster and more secure.

Example: VMware ESX, Microsoft Hyper-V.

Type 2 Hypervisor: Runs on top of an operating system. 5
mainly used for testing.

Example: Oracle Virtual Box, VMware Workstation.

3. Role of APIs in cloud service integration:

APIs allow different cloud services to communicate.

They enable data sharing and application integration.

APIs support automation and remote access. They improve interoperability between different cloud providers.

4) Resource Pooling in cloud computing.

* Resource pooling means servers, storage, and networks are shared among many users.

* Resources are allocated based on user demand.

* It improves resource utilization.

* It reduces cost and supports multiple users.

efficiently.

5) Rapid Elasticity in Cloud Computing.

Rapid elasticity is the ability to quickly increase or decrease resources.

It automatically adjusts to workload changes.

It ensures good performance during high demand.

It helps reduce costs by using only the required resources.